

Hard level programs

The screenshot shows the Programiz Online Java Compiler interface. The code in `Main.java` is a tax calculation program. It prompts the user to enter income, calculates taxable income and tax based on brackets, and prints the results. The output shows the calculation for an income of 5600.

```
3 public class Main {
4     public static void main(String[] args) {
18         if (income <= 250000) {
20             tax = 0;
21         }
22         else if (income <= 500000) {
23             taxableIncome = income - 250000;
24             tax = taxableIncome * 0.10;
25         }
26         else if (income <= 1000000) {
27             taxableIncome = income - 250000;
28             tax = taxableIncome * 0.20;
29         }
30         else {
31             taxableIncome = income - 250000;
32             tax = taxableIncome * 0.30;
33         }
34
35         System.out.println("Taxable Income: " + taxableIncome);
36         System.out.println("Tax = " + tax);
37     }
38 }
```

Output:

```
Enter the income: 5600
Taxable Income: 0.0
Tax = 0.0

=== Code Execution Successful ===
```

The screenshot shows the Programiz Online Java Compiler interface. The code in `Main.java` is a program to calculate a student's class based on marks in various subjects. It prompts the user to enter marks in Python, C Programming, Mathematics, Physics, Chemistry, and Professional Ethics, calculates the total and average, and prints the class. The output shows the calculation for the given marks.

```
11 class Student implements Marks, Result {
38     public void calculate() {
49         if (aggregate > 75)
50             System.out.println("Class: DISTINCTION");
51         else if (aggregate >= 60)
52             System.out.println("Class: FIRST DIVISION");
53         else if (aggregate >= 50)
54             System.out.println("Class: SECOND DIVISION");
55         else if (aggregate >= 40)
56             System.out.println("Class: THIRD DIVISION");
57         else
58             System.out.println("Class: FAIL");
59     }
60 }
61
62 public class Main {
63     public static void main(String[] args) {
64         Student s = new Student();
65         s.getMarks();
66         s.calculate();
67     }
68 }
```

Output:

```
Enter marks in Python: 67
Enter marks in C Programming: 78
Enter marks in Mathematics: 89
Enter marks in Physics: 56
Enter marks in Chemistry: 67
Enter marks in Professional Ethics: 45
Total = 402.0
Aggregate = 67.00
Class: FIRST DIVISION

=== Code Execution Successful ===
```

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Main.java Run

```
3 public class Main {
4     public static void main(String[] args) {
5         // ...
6
7         if (n < 0) {
8             System.out.println("Invalid input");
9             return;
10        }
11
12        int steps = 0;
13
14        while (n != 0) {
15            if (n % 2 == 0)
16                n = n / 2;
17            else
18                n = n - 1;
19            steps++;
20        }
21
22        System.out.println("Number of steps = " + steps);
23    }
24 }
```

Output

Enter n: 6
Number of steps = 4

=== Code Execution Successful ===

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Main.java Run

```
3 public class Main {
4     public static void main(String[] args) {
5         // ...
6
7         Arrays.sort(rows, (a, b) -> {
8             for (int x : mat[b])
9                 if (countA != countB)
10                     return countA - countB;
11             return a - b;
12         });
13
14         System.out.print("[");
15
16         for (int i = 0; i < k && i < m; i++) {
17             System.out.print(rows[i]);
18
19             if (i < k - 1 && i < m - 1)
20                 System.out.print(", ");
21         }
22
23         System.out.println("]");
24     }
25 }
```

Output

8
9
Enter row 3:
1
2
2
3
Enter row 4:
4
5
6
7
Enter row 5:
7
8
9
0
Enter k: 2
[3, 4]

=== Code Execution Successful ===

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Main.java Run

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter ransomNote: ");
8         String ransom = sc.nextLine();
9
10        System.out.print("Enter magazine: ");
11        String magazine = sc.nextLine();
12
13        int[] count = new int[256];
14
15        for (char c : magazine.toCharArray())
16            count[c]++;
17
18        boolean possible = true;
19
20        for (char c : ransom.toCharArray()) {
21            if (count[c] == 0) {
22                possible = false;
23                break;
24            }
25        }
26    }
27 }
```

Output

```
Enter ransomNote: likitha
Enter magazine: likki
false

=== Code Execution Successful ===
```

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Main.java Run

```
3 public class Main {
17     public static void main(String[] args) {
24         Scanner sc = new Scanner(System.in);
25         String s = sc.nextLine();
26         int current = value(s.charAt(0));
27
28         if (current == 0) {
29             System.out.println("Invalid Roman numeral");
30             return;
31         }
32
33         if (i + 1 < s.length() &&
34             current < value(s.charAt(i + 1))) {
35             result -= current;
36         } else {
37             result += current;
38         }
39     }
40
41     System.out.println(result);
42 }
43 }
```

Output

```
Enter Roman numeral: 6
Invalid Roman numeral

=== Code Execution Successful ===
```

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Main.java

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter string: ");
8         String s = sc.nextLine();
9
10        s = s.trim();
11
12        int index = s.lastIndexOf(' ');
13
14        if (index == -1)
15            System.out.println(s.length());
16        else
17            System.out.println(s.length() - index - 1);
18    }
19 }
```

Run

Output

```
Enter string: rama
4
=== Code Execution Successful ===
```

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Main.java

```
3 public class Main {
4     public static void main(String[] args) {
5
6         int[] b = new int[m];
7
8         System.out.println("Enter second array:");
9         for (int i = 0; i < m; i++)
10            b[i] = sc.nextInt();
11
12        ArrayList<Integer> result = new ArrayList<>();
13
14        for (int x : a) {
15            for (int y : b) {
16                if (x == y && !result.contains(x)) {
17                    result.add(x);
18                    break;
19                }
20            }
21        }
22
23        System.out.println(result);
24    }
25 }
```

Run

Output

```
Enter size of first array: 7
Enter first array:
4
7
8
9
0
5
7
Enter size of second array: 2
Enter second array:
5
6
[5]
=== Code Execution Successful ===
```

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Main.java Run

```
3 public class Main {
4     public static void main(String[] args) {
5
6
7
8
9
10
11
12
13
14
15         System.out.print("[");
16
17         for (int i = 1; i <= n; i++) {
18             if (i % 15 == 0)
19                 System.out.print("\nFizzBuzz");
20             else if (i % 3 == 0)
21                 System.out.print("\nFizz");
22             else if (i % 5 == 0)
23                 System.out.print("\nBuzz");
24             else
25                 System.out.print("\n" + i + "\n");
26
27             if (i < n)
28                 System.out.print(", ");
29         }
30
31         System.out.println("]");
32     }
33 }
```

Output

Enter n: 7
["1", "2", "Fizz", "4", "Buzz", "Fizz", "7"]
=== Code Execution Successful ===

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Main.java Run

```
3 public class Main {
4     public static void main(String[] args) {
5         Thread t2 = new Thread() -> {
6             try {
7                 Thread.sleep(1000);
8             } catch (InterruptedException e) {
9                 Thread.currentThread().interrupt();
10            }
11
12            account.deposit(deposit);
13        });
14
15        t1.start();
16        t2.start();
17
18        try {
19            t1.join();
20            t2.join();
21        } catch (InterruptedException e) {
22            Thread.currentThread().interrupt();
23        }
24    }
25 }
```

Output

Enter withdrawal amount: 4000
Enter deposit amount: 499
Withdraw requested: 4000
Withdraw operation success
Balance amount: 6000
Deposited amount: 499
New balance: 6499
=== Code Execution Successful ===

Main.java Run

```
3 public class Main {
4     static long reverse(long n) {
5         long rev = 0;
6
7         while (n > 0) {
8             rev = rev * 10 + n % 10;
9             n /= 10;
10        }
11
12        return rev;
13    }
14
15    static boolean isPalindrome(long n) {
16        return n == reverse(n);
17    }
18
19    public static void main(String[] args) {
20        Scanner sc = new Scanner(System.in);
21
22        System.out.print("Enter number: ");
23        long n = sc.nextLong();
```

Output

Enter number: 2345
2345 + 5432 = 7777
7777 is a Palindrome
=== Code Execution Successful ==

Main.java

Run

Output

```
3 public class Main {
5     static class BankAccount {
6     }
7
48 }
49
50 public static void main(String[] args) {
51     Scanner sc = new Scanner(System.in);
52
53     BankAccount account = new BankAccount();
54
55     account.readDetails(sc);
56
57     System.out.print("Enter deposit amount: ");
58     double deposit = sc.nextDouble();
59     account.deposit(deposit);
60
61     System.out.print("Enter withdrawal amount: ");
62     double withdraw = sc.nextDouble();
63     account.withdraw(withdraw);
64
65     account.display();
66 }
67 }
```

```
Account Number: 192425
Depositor Name: 678
Account Type: 98
Enter deposit amount: 098
Amount deposited: 98.0
Enter withdrawal amount: 4000
Amount withdrawn: 4000.0
Account Number: 192425
Name: 678
Account Type: 98
Balance: 6098.0
```

=== Code Execution Successful ===

Main.java

```
3 public class Main {
4     public static void main(String[] args) {
5
10         System.out.print("Enter upper range: ");
11         int high = sc.nextInt();
12
13         if (low > high) {
14             System.out.println("[ ]");
15             return;
16         }
17
18         System.out.print("[ ");
19
20         for (int i = low; i <= high; i++) {
21             System.out.print("(" + i + ", " + (i * i) + ")");
22
23             if (i < high)
24                 System.out.print(", ");
25         }
26
27         System.out.println("]");
28     }
29 }
```

Main.java	Run	Output
<pre>3 public class Main { 4 static void permute(char[] arr, int index) { 14 for (int i = index; i < arr.length; i++) { 27 } 28 } 29 30 public static void main(String[] args) { 31 Scanner sc = new Scanner(System.in); 32 33 System.out.print("Given Number: "); 34 String s = sc.next(); 35 36 if (s.startsWith("-")) 37 s = s.substring(1); 38 39 System.out.println("Permutations are:"); 40 41 char[] arr = s.toCharArray(); 42 Arrays.sort(arr); 43 44 permute(arr, 0); 45 } }</pre>		Given Number: 6 Permutations are: 6 === Code Execution Successful

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Main.java	Run	Output
<pre>3 public class Main { 15 public static void main(String[] args) { 21 System.out.print("Enter upper range: "); 22 int high = sc.nextInt(); 23 24 if (low > high high < 0) { 25 System.out.println(""); 26 return; 27 } 28 29 ArrayList<Integer> result = new ArrayList<>(); 30 31 for (int i = Math.max(0, low); i <= high; i++) { 32 int root = (int) Math.sqrt(i); 33 34 if (root * root == i && digitSum(i) < 10) 35 result.add(i); 36 } 37 38 System.out.println(result); 39 } 40 }</pre>		Enter lower range: 6 Enter upper range: 8 [] === Code Execution Successful

Main.javaRun

```
3 public class Main {
5     public static void main(String[] args) {
7
9         System.out.println(n + "th Prime number is " + nthPrime);
10
11         System.out.print(n + " prime numbers after " + nthPrime + " are: ");
12
13         count = 0;
14         number = nthPrime + 1;
15
16         while (count < n) {
17             if (isPrime(number)) {
18                 System.out.print(number);
19                 count++;
20
21                 if (count < n)
22                     System.out.print(", ");
23             }
24             number++;
25         }
26     }
27 }
```

Output

```
N = 7
7th Prime number is 17
7 prime numbers after 17 are: 19, 23, 29, 31, 37, 41, 43
=== Code Execution Successful ===
```

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Main.javaRun

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Given Number: ");
8         int n = sc.nextInt();
9
10        System.out.print("N = ");
11        int k = sc.nextInt();
12
13        if (n <= 0 || k <= 0) {
14            System.out.println("Invalid input");
15            return;
16        }
17
18        ArrayList<Integer> factors = new ArrayList<>();
19
20        for (int i = 1; i <= n; i++) {
21            if (n % i == 0)
22                factors.add(i);
23        }
24    }
25 }
```

Output

```
Given Number: 75
N = 6
Number of factors = 6
6th factor of 75 = 75
=== Code Execution Successful ===
```

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Main.java Run

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Total Users: ");
8         int total = sc.nextInt();
9
10        System.out.print("Staff Users: ");
11        int staff = sc.nextInt();
12
13        int nonTeaching = staff / 3;
14        int student = total - staff - nonTeaching;
15
16        if (student < 0)
17            student = 0;
18
19        System.out.println("Student Users: " + student);
20    }
21 }
```

Output

```
Total Users: 5
Staff Users: 4
Student Users: 0

=== Code Execution Successful ===
```

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Main.java Run

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter the number of days: ");
8         int days = sc.nextInt();
9
10        int years = days / 365;
11        int remaining = days % 365;
12        int weeks = remaining / 7;
13        int d = remaining % 7;
14
15        System.out.println("No. of years: " + years);
16        System.out.println("No. of weeks: " + weeks);
17        System.out.println("No. of days: " + d);
18    }
19 }
```

Output

```
Enter the number of days: 6
No. of years: 0
No. of weeks: 0
No. of days: 6

=== Code Execution Successful ===
```

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Main.java Run

```

3 public class Main {
4     public static void main(String[] args) {
26
27
28         double bill;
29
30         if (units <= 100) {
31             bill = units * 1;
32         } else if (units <= 200) {
33             bill = 100 + (units - 100) * 2.5;
34         } else if (units <= 500) {
35             bill = 100 + 100 * 2.5 + (units - 200) * 4;
36         } else {
37             bill = 100 + 100 * 2.5 + 300 * 4 + (units - 500) * 6;
38         }
39
40         System.out.println("\nConsumer No: " + no);
41         System.out.println("Consumer Name: " + name);
42         System.out.println("Units Consumed: " + units);
43         System.out.println("Bill Amount: Rs." + bill);
44     }
45 }

```

Output

```

Consumer No: 67
Consumer Name: roshitha
Previous Month Reading: 678
Current Month Reading: 890

Consumer No: 67
Consumer Name: roshitha
Units Consumed: 212
Bill Amount: Rs.398.0

=== Code Execution Successful ===

```

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Main.java Run

```

46 class Motorcycle extends Vehicle {
54     void displayMotorcycle() {
57     }
58 }
59
60 public class Main {
61     public static void main(String[] args) {
62
63         Car car = new Car("TN01AB1234", "Ravi", 4);
64         Truck truck = new Truck("TN02CD5678", "Raju", 10);
65         Motorcycle bike = new Motorcycle("TN03EF9999", "Arun", 150);
66
67         System.out.println("CAR");
68         car.displayCar();
69
70         System.out.println("\nTRUCK");
71         truck.displayTruck();
72
73         System.out.println("\nMOTORCYCLE");
74         bike.displayMotorcycle();
75     }
76 }

```

Output

```

CAR
License Plate: TN01AB1234
Owner: Ravi
Number of Doors: 4

TRUCK
License Plate: TN02CD5678
Owner: Raju
Capacity: 10.0 tons

MOTORCYCLE
License Plate: TN03EF9999
Owner: Arun
Engine: 150 CC

=== Code Execution Successful ===

```

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Main.java

Run

Output

```
22 public class Main {
23     public static void main(String[] args) {
24         // Create a Scanner object to read input
25         Scanner sc = new Scanner(System.in);
26
27         // Create a Person object
28         Person p = new Person();
29         p.name = sc.nextLine();
30
31         System.out.print("Address: ");
32         p.address = sc.nextLine();
33
34         System.out.print("Telephone Number: ");
35         p.telephone = sc.nextLine();
36
37         System.out.print("Mobile Number: ");
38         p.mobile = sc.nextLine();
39
40         System.out.print("Head of Family: ");
41         p.head = sc.nextLine();
42
43         System.out.print("Unique ID: ");
44         p.uniqueId = sc.nextLine();
45
46         p.display();
47     }
48 }
```

Name: roshitha
Address: pakala
Telephone Number: 9125026814
Mobile Number: 78904563
Head of Family: u r naidu
Unique ID: 678

--- Person Details ---
Name: roshitha
Address: pakala
Telephone: 9125026814
Mobile: 78904563
Head of Family: u r naidu
Unique ID: 678

=== Code Execution Successful ===

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Main.java Run

```
3 public class Main {
5     static int minCuts(String s) {
21         for (int i = 0; i < n; i++) {
33             }
34         }
35         return dp[n - 1];
36     }
37 }
38 public static void main(String[] args) {
39     Scanner sc = new Scanner(System.in);
40 }
41 System.out.print("Enter string: ");
42 String s = sc.nextLine();
43 }
44 if (s.length() == 0) {
45     System.out.println(0);
46     return;
47 }
48 }
49 System.out.println("Minimum cuts = " + minCuts(s));
50 }
51 }
```

Output

Enter string: radha
Minimum cuts = 4

=== Code Execution Successful ===

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Main.java Run

```
3 public class Main {
59 }
60 public static void main(String[] args) {
61     Scanner sc = new Scanner(System.in);
62 }
63 System.out.print("Enter first number: ");
64 int a = sc.nextInt();
65 }
66 System.out.print("Enter second number: ");
67 int b = sc.nextInt();
68 }
69 Thread t1 = new Addition(a, b);
70 Thread t2 = new Subtraction(a, b);
71 Thread t3 = new Multiplication(a, b);
72 Thread t4 = new Division(a, b);
73 }
74 t1.start();
75 t2.start();
76 t3.start();
77 t4.start();
78 }
79 }
```

Output

Enter first number: 567
Enter second number: 789
Addition = 1356
Multiplication = 447363
Subtraction = -222
Division = 0.7186311787072244

=== Code Execution Successful ===

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Main.java Run

```
1 public class Main {
2
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5
6         System.out.print("Enter first number: ");
7         int a = sc.nextInt();
8
9         System.out.print("Enter second number: ");
10        int b = sc.nextInt();
11
12        Thread t1 = new Addition(a, b);
13        Thread t2 = new Subtraction(a, b);
14        Thread t3 = new Multiplication(a, b);
15        Thread t4 = new Division(a, b);
16
17        t1.start();
18        t2.start();
19        t3.start();
20        t4.start();
21    }
22 }
```

Output

```
Enter first number: 567
Enter second number: 789
Addition = 1356
Multiplication = 447363
Subtraction = -222
Division = 0.7186311787072244
=== Code Execution Successful ===
```

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Main.java Run

```
1 public class Main {
2
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5
6         System.out.print("Number of words: ");
7         int n = sc.nextInt();
8
9         String[] words = new String[n];
10
11        for (int i = 0; i < n; i++)
12            words[i] = sc.next();
13
14        WordFilter wf = new WordFilter(words);
15
16        System.out.print("Prefix: ");
17        String pref = sc.next();
18
19        System.out.print("Suffix: ");
20        String suff = sc.next();
21
22        System.out.println("Index = " + wf.f(pref, suff));
23    }
24 }
```

Output

```
Number of words: 5
ro
sh
it
ha
upp
Prefix: ro
Suffix: fh
Index = -1
=== Code Execution Successful ===
```

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Main.java **Run**

```

3 public class Main {
26     public static void main(String[] args) {
41         System.out.print("Enter left index: ");
42         int left = sc.nextInt();
43
44         System.out.print("Enter right index: ");
45         int right = sc.nextInt();
46
47         System.out.println("Sum = " + obj.sumRange(left, right));
48
49         System.out.print("Enter index to update: ");
50         int index = sc.nextInt();
51
52         System.out.print("Enter new value: ");
53         int value = sc.nextInt();
54
55         obj.update(index, value);
56
57         System.out.println("Updated sum = " +
58                             obj.sumRange(left, right));
59     }
60 }

```

Output

```

Enter array size: 4
Enter array:
6
7
8
9
Enter left index: 2
Enter right index: 7
ERROR!
Exception in thread "main" java.lang.ArrayIndexOutOfBoundsException: Index 4 out
of bounds for length 4
    at Main$NumArray.sumRange(Main.java:20)
    at Main.main(Main.java:47)

=== Code Exited With Errors ===

```

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Main.java **Run**

```

3 public class Main {
4     public static void main(String[] args) {
12         for (int i = 0; i < n; i++)
13             arr[i] = sc.nextInt();
14
15         Set<Integer> all = new HashSet<>();
16         Set<Integer> current = new HashSet<>();
17
18         for (int x : arr) {
19             Set<Integer> next = new HashSet<>();
20
21             next.add(x);
22
23             for (int value : current)
24                 next.add(value | x);
25
26             current = next;
27             all.addAll(current);
28         }
29
30         System.out.println("Number of distinct ORs = " + all.size());
31     }
32 }

```

Output

```

Enter array size: 3
2
4
5
Number of distinct ORs = 5

=== Code Execution Successful ===

```

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Main.java

Run

Output

```
3 public class Main {
4     public static void main(String[] args) {
5         // Enter size of first array: 3
6         int n = sc.nextInt();
7
8         int[] a = new int[n];
9
10        for (int i = 0; i < n; i++)
11            a[i] = sc.nextInt();
12
13        System.out.print("Enter size of second array: ");
14        int m = sc.nextInt();
15
16        int[] b = new int[m];
17
18        for (int i = 0; i < m; i++)
19            b[i] = sc.nextInt();
20
21        int[] result = addNegabinary(a, b);
22
23        System.out.println(Arrays.toString(result));
24    }
25 }
```

```
Enter size of first array: 3
2
4
5
Enter size of second array: 6
7
8
9
2
3
5
[1, 0, 1, 0, 0, 1, 0, 0]

=== Code Execution Successful ===
```

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Main.java

Run

Output

```
3 public class Main {
4     public static void main(String[] args) {
5         // Enter array size: 3
6         for (int i = 0; i < n; i++)
7             arr[i] = sc.nextInt();
8
9         Set<Integer> all = new HashSet<>();
10        Set<Integer> current = new HashSet<>();
11
12        for (int x : arr) {
13            Set<Integer> next = new HashSet<>();
14
15            next.add(x);
16
17            for (int value : current)
18                next.add(value | x);
19
20            current = next;
21            all.addAll(current);
22        }
23
24        System.out.println("Number of distinct ORs = " + all.size());
25    }
26 }
```

```
Enter array size: 3
2
4
5
Number of distinct ORs = 5

=== Code Execution Successful ===
```